

Data Processing with petropandas

Module 1 — Practical Thermodynamic Modelling for Petrology

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Introduction to Metamorphic Petrology

What Is Metamorphic Petrology?

- Metamorphism: the mineralogical and textural response of a rock to changing pressure, temperature, and fluid/melt composition — without wholesale melting
- Metamorphic petrology reads that response to reconstruct a rock's **P-T-t (-X) history**
- That history records tectonic processes we cannot watch directly: burial, subduction, collision, exhumation

Central question

Given the minerals we see today, what pressures and temperatures did this rock pass through?

How Do We Study Metamorphic Rocks?

1. **Field:** sampling, structural context, outcrop-scale relationships
2. **Petrography:** thin-section microscopy — mineral assemblage, textures, reaction relationships, deformation fabric
3. **Mineral chemistry:** quantitative composition of each phase — needed to turn a qualitative textural story into a quantitative P-T estimate
4. **Thermodynamic modelling:** today's topic — turn (3) into P-T constraints (Modules 2 & 3)

Getting Mineral Chemistry: the Electron Microprobe

- **EPMA** (electron probe microanalysis): a focused electron beam ($\sim 1\text{--}5\ \mu\text{m}$) excites characteristic X-rays, measured by wavelength-dispersive spectrometers (WDS) against calibrated standards
- Output: a quantitative **oxide wt% spot analysis** for one point in one mineral grain — exactly the row format `petropandas` expects
- **SEM-EDS** area scans / element maps are faster and less precise, but useful for texture-guided sampling before EPMA, or for semi-quantitative area-integrated compositions

The Problem of Local Equilibrium

Thermodynamic modelling assumes...

...that every component in the system had the chance to equilibrate together, at the same P - T , in the same reactive volume.

But real rocks are heterogeneous at the mm-cm scale:

- Zoned porphyroblasts preserve growth history, not equilibrium
- Matrix, inclusions, and retrograde rims may never have shared one equilibrium
- Veins, segregations, and weathering add material unrelated to the reaction of interest

A hand-sample-scale **whole-rock (XRF/ICP) bulk composition** averages straight across all of this.

Why Point / Area Analyses Are More Suitable

- An **EPMA spot analysis** of matrix phases in mutual textural contact — or an **EDS area analysis** restricted to one textural domain — samples only the volume that plausibly equilibrated
- Averaging those spot/area analyses gives an **effective (local) bulk composition**: the right input for a pseudosection, not a whole-rock number diluted by unrelated material
- Whole-rock XRF still has its place — protolith classification, igneous petrogenesis — just not as the default input for metamorphic phase-equilibrium modelling

From Sample to petropandas

What comes off the microprobe: one CSV row per analysed spot, oxide wt% columns, mineral/sample labels.

petropandas takes that file exactly as-is and turns it into APFU, end-members, and an effective bulk composition — ready for Modules 2 & 3.

Mineral	SiO2	Al2O3	FeO	MgO
Grt	37.9	21.4	28.1	3.2
Grt	38.1	21.3	27.6	3.6
Bt	35.2	18.9	19.8	9.1

Why Scripting & Reproducibility?

Spreadsheet-Based Petrology: Hidden Risks

The Traditional Excel Workflow

- Copy-pasting raw EPMA data into complex calculation spreadsheets
- Separate sheets or workbooks for each mineral group or sample
- Manual copy-pasting to format inputs for thermodynamic software

Common Pitfalls

- **Opaque calculations:** Hidden, broken, or inconsistent cell formulas across sheets
- **Data corruption:** Accidental cell overwrites with zero audit trail
- **Non-scalable:** Processing 1,000s of spot analyses becomes unmaintainable

"Spreadsheet rot" makes published geochemical recalculations difficult to audit and reproduce.

The Scripting Advantage: Workflows over Worksheets

- **Explicit & Transparent:** Every calculation step (unit conversions, oxygen basis, $\text{Fe}^{3+}/\text{Fe}^{2+}$ split) is written in code, readable, and inspectable
- **Scalable & Automated:** Process 10 or 10,000 EPMA spot analyses with the exact same 3 lines of code
- **Pristine Raw Data:** Raw EPMA CSV files are never edited directly — all transformations occur programmatically in memory
- **Direct Data Pipeline:** Seamless link from raw microprobe CSV → mineral APFU → bulk composition → thermodynamic input files

Key Mindset Shift

Instead of managing static spreadsheet files, you build an **automated, executable data pipeline**.

Reproducible Research in Metamorphic Petrology

- **Executable Documentation:** Jupyter Notebooks integrate raw data ingestion, stoichiometric calculations, publication figures, and text in one place
- **FAIR Geochemistry:** Open, re-runnable scripts allow reviewers and readers to inspect and verify how compositions were derived
- **Future-Proof Workflows:**
 - Re-run analyses in seconds when microprobe calibrations or standards update
 - Re-calculate mineral end-members instantly as solution models ($a-x$) evolve
- **Version Control:** Track every modification with Git — no more `Garnet_recalc_v2_FINAL_fixed.xlsx`

Reproducibility Principle

A P-T pseudosection is only as reliable as the mineral recalculations and bulk compositions that feed into it.

Objective

Process raw microprobe and whole-rock measurements into thermodynamic-ready compositions using `petropandas`.

- Data ingestion (EPMA spot analyses, XRF whole-rock data)
- Mineral recalculations to APFU and end-members
- Whole-rock geochemistry & thermodynamic bulk formatting
- Petrological plots

What is petropandas?

What is petropandas?

Elevator pitch

A **pandas-accessor library** for electron-microprobe (EPMA) mineral analyses and whole-rock geochemistry: oxide wt% → moles → atoms-per-formula-unit (APFU) → site-allocated structural formulas → end-members, plus bulk-rock helpers and plotting.

Everything hangs off DataFrame accessors:

```
df.oxides  df.moles  df.cations  df.mineral  df.bulk
```

No new file formats, no new data structures — just plain pandas.

Installation

Requires Python ≥ 3.12 . Installed this morning as part of one shared uv project together with pypsbuilder (Module 2):

```
uv add petropandas[lab] pypsbuilder[pyqt6]
```

Run notebooks with `uv run jupyter lab`. See `install.md` in the workshop repository for the from-scratch/downloaded-materials options and full details.

Core Workflow

Accessor	Purpose
<code>df.oxides</code>	ingest & standardize raw oxide wt% data
<code>df.moles</code> / <code>df.cations</code>	unit conversion (wt% $\leftrightarrow$ moles $\leftrightarrow$ APFU)
<code>df.mineral</code>	structural formulas, end-members, QC
<code>df.bulk</code>	whole-rock / bulk-rock geochemistry
<code>plotting</code>	ScatterPlot, TernaryPlot, ProfilePlot

1. Data Ingestion — `df.oxides`

- Auto-detects and standardizes oxide columns from CSV (SiO_2 , Al_2O_3 , FeO , ...)
- Reorders columns to petrological convention
- Normalizes analyses to 100%
- Splits total Fe into Fe^{3+} / Fe^{2+} (Droop or Schumacher method)
- Select / filter / mean analyses by group (e.g. by mineral, by sample)

2. Unit Conversion — `df.moles`, `df.cations`

- Chainable, reversible conversions: `wt% ↔ moles ↔ atoms-per-formula-unit`
- Current unit tracked automatically via `df.attrs["petro_units"]`
- No manual bookkeeping of oxide molar masses or cation charges

3. Mineral Recalculation — `df.mineral`

- APFU on the correct oxygen basis, with valence splitting
- Site-allocated structural formulas
- End-member proportions (e.g. Prp, Alm, Sps, Grs for garnet)
- 0–1 stoichiometry quality-control scores

Coverage

16 IMA mineral models (garnet, feldspar, cpx/opx, mica, amphibole, ...) plus 33 THERMOCALC a-x solution models via `petropandas.hpxeos` — ready for `pypsbuilder` / `MAGEMin`.

4. Whole-Rock Geochemistry — df.bulk

- Weighted mean bulk compositions
- CIPW norm
- Alumina saturation (A/NK, A/CNK, Shand classification)
- Oxide ratios: Mg#, FeO_T , K/Na, ...
- **Formatted bulk-composition output** for THERMOCALC, Perple_X, and MAGEMin — the direct link to Modules 2 & 3

Live Code: Oxides → End-Members

```
from petropandas import pd, Grt
from petropandas.data import minerals as df

# Select garnet analyses, sort columns petrologically
g = df.oxides.select("Garnet", on="Mineral").oxides.sorted()

# APFU on a 12-oxygen basis, Fe3+/Fe2+ (Droop) splitting
g.mineral.apfu(Grt)

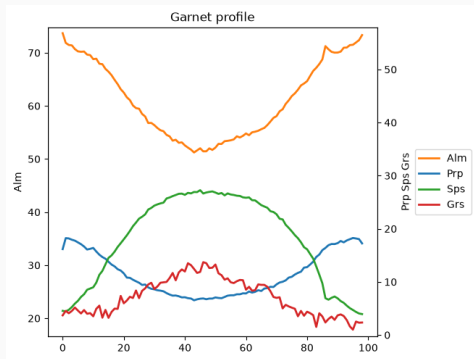
# Site-allocated structural formula
g.mineral.site_allocations(Grt)

# End-member proportions (Prp, Alm, Sps, Grs, ... )
g.mineral.end_members(Grt)

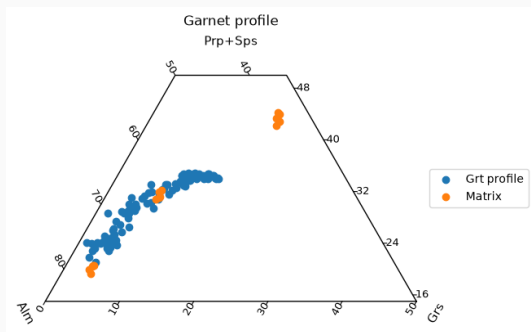
# QC: per-criterion stoichiometry scores (0-1)
g.mineral.check_stoichiometry(Grt)
```

Plotting

Petrological Plots



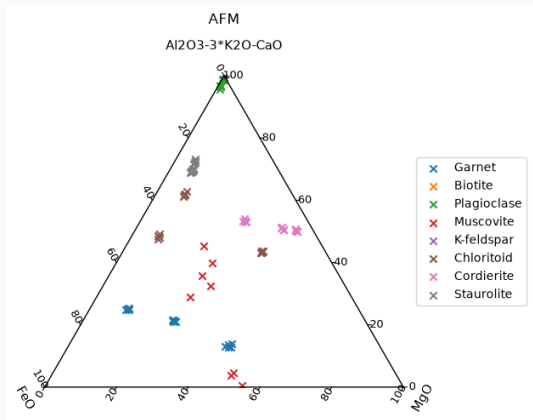
Garnet compositional profile



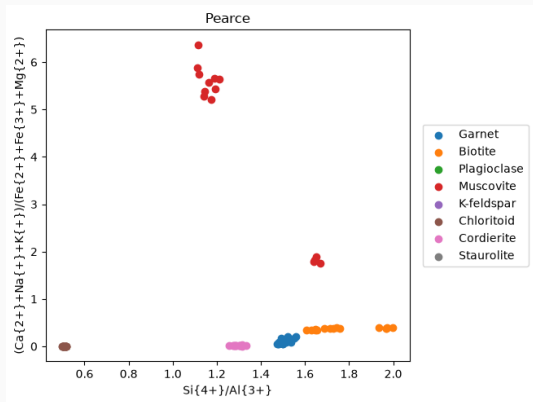
Garnet ternary: profile vs. matrix

ScatterPlot / TernaryPlot / ProfilePlot use `pandas.eval()` expressions directly as plot axes.

More Plot Examples



AFM ternary across 8 mineral phases



Pearce-style discrimination scatter plot

Hands-on

Hands-on Exercise

1. Load your own (or the provided example) EPMA CSV with `df.oxides`
2. Select one mineral group and compute APFU + end-members
3. Run `check_stoichiometry` — flag any low-quality analyses
4. Build a bulk-rock composition with `df.bulk` and export it in **THERMOCALC** format
5. Make one `ScatterPlot` or `ProfilePlot` of your choice

Bridge to Module 2

The THERMOCALC-formatted bulk composition you export here is exactly what `pypsbuilder` needs as input.

Recap

Recap & Resources

Key takeaways

- Raw oxide data → APFU → end-members → bulk composition, all in pandas
- Built-in stoichiometry QC catches bad analyses early
- Direct export to THERMOCALC / Perple_X / MAGEMin formats

Resources:

- Documentation: petropandas docs (walkthrough notebook)
- Source: github.com/ondrolexa/petropandas

Next: P-T Pseudosections with pypsbuilder